Unit 11 - Task 'Development Individual Project: Executive Summary and Assessment' - AI and Cloud Computing

Cloud Strategy Assessment Report for CoinMeister: Designing a Secure and Scalable FinTech Cloud Platform

Executive Summary

CoinMeister is a hypothetical FinTech company providing digital payment processing and digital wallet services. As transaction volumes increase, Legacy infrastructure limits scalability, increases costs and restricts access to modern FinTech capabilities. This report recommends a hybrid cloud strategy combining which consists of private segmented cloud infrastructure for highly sensitive financial workloads with public cloud services for applications, data analytics and artificial intelligence (AI). This strategy incorporates Infrastructure as Code (IaC), automation, serverless computing and advanced security controls. The proposed cloud approach will improve transaction scalability, reduce infrastructure costs, enhance defense capabilities using advanced fraud detection and strengthen business resilience. It will also improve compliance capabilities in a highly regulated environment characterized by diverse international regulatory requirements, through effective governance, monitoring, strong identity and access management and by adopting this strategy, CoinMeister can excel in the creation of a secure and flexible platform capable of supporting future FinTech innovation.

(Mather, et al. 2009)

1. Introduction

CoinMeister is a hypothetical FinTech company providing digital payment processing and digital wallet services for consumers and businesses. As transaction volumes continue to grow, its legacy infrastructure struggles to scale efficiently, resulting in higher operational costs, slower service deployment and increased compliance challenges. Cloud computing provides the scalability, flexibility and advanced technologies required to support secure, resilient and compliant financial services. Hybrid cloud strategies are increasingly being adopted because they allow organizations to integrate operational flexibility with stronger privacy control over critical workloads.

(Anand, 2025).

2. Cloud Solution Design

2.1 Hybrid Cloud Architecture

A hybrid cloud architecture is recommended for CoinMeister because it combines the security advantages of private cloud deployment with the scalability of public cloud services, such approach helps promote the benefits of workload availability to be placed according to security, performance and compliance requirements.

(Anand, 2025).

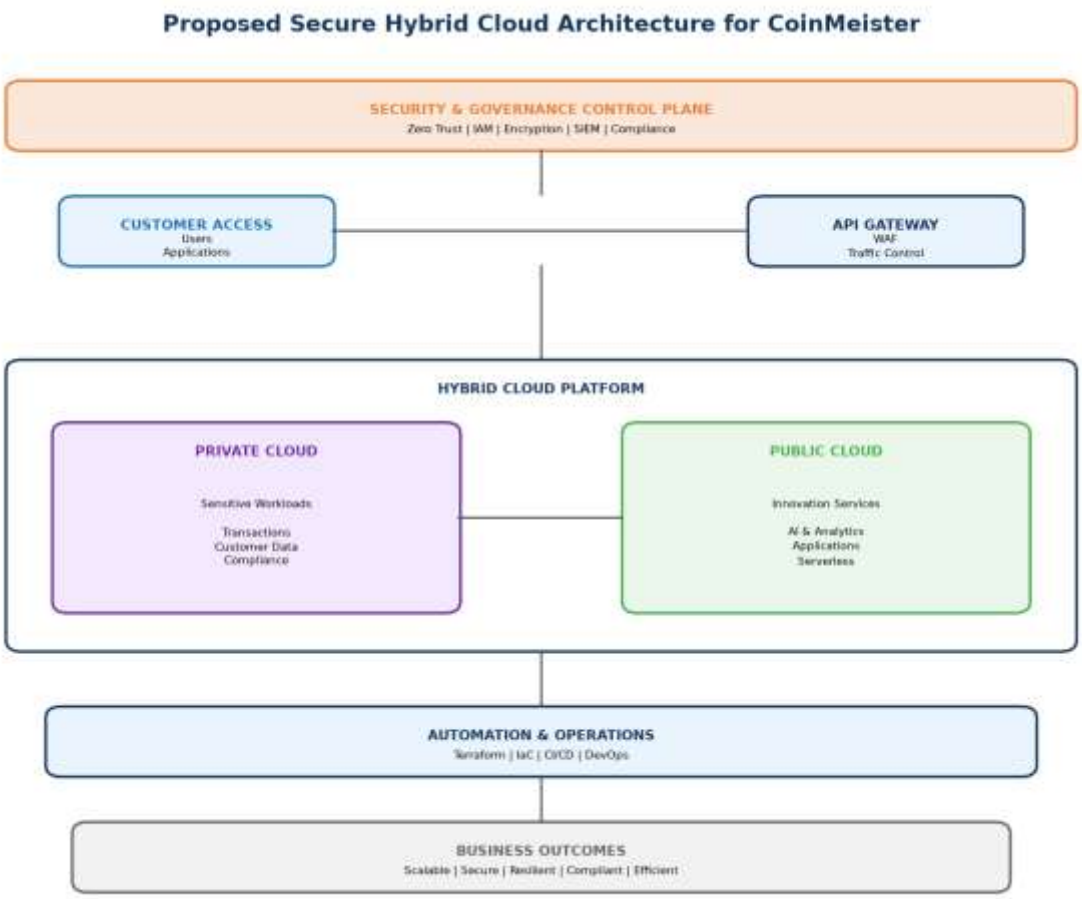
Unlike in the public cloud, private clouds could provide organizations with a more flexible, controlled granular access controls implementation, where stricter security measures, and dedicated governance processes can be tailored to meet specific risk requirements. This helps to increase the control of privacy and enables organizations to swiftly restrict or suspend access, when necessary, with minimal delay, thus limiting

potential exposure and controlling authorized and unauthorized access through stronger isolation and segmentation of critical assets, providing a stricter controlled security boundary where sensitive information can be hosted and protected with greater precision. In addition to more control of operational and usage data collected by third-party cloud providers. On the other hand, public cloud resources should support customer applications, analytics platforms, AI services and application development environments. The utilization of public cloud platforms could provide a more elastic and dynamic computing capacity that could be increased during periods of high transaction activity and be reduced when demand decreases. Using this strategy would support improved compliance while enabling better innovation. Sensitive workloads would remain controlled in a more fine-grained set of preferences, while public cloud services provide scalable computing and advanced technologies. Security must be applied consistently across both environments. Encryption, authentication and monitoring controls are essential for protecting confidentiality, integrity and availability of data.

(Mather, et al. 2009).

The proposed hybrid cloud architecture for CoinMeister is illustrated in Figure 1. The design separates sensitive financial workloads from scalable public cloud services, enabling the organization to maintain regulatory compliance while benefiting from cloud elasticity, automation and advanced technologies.

Figure 1: Proposed Hybrid Cloud Architecture for CoinMeister



2.2 Infrastructure as Code Implementation

In order to automate cloud resource management, it is highly advised that CoinMeister, implements Infrastructure as Code (IaC). Terraform is one solution that provides a cloud-agnostic IaC approach and enables effective and consistent infrastructure deployment through computer code. Furthermore, Terraform uses cloud-specific providers to interact with platforms such as Azure, AWS or Google Cloud Platform (GCP), which are leading vendors. The trade-off to consider tough is the possibility that some configurations could remain provider-dependent, in this case HasiCorp's

Terraform. Infrastructure configurations are being stored under strict version control and deployed consistently, thus reducing amount of configuration errors and improving deployment efficiency. Automated infrastructure management, administration effort cost and supports DevOps practices by integrating infrastructure changes with CI/CD pipelines. Version control supports stronger governance because infrastructure changes can now be reviewed, tracked and audited, this is particularly useful for organizations operating in financial regulated environments. Using Terraform alongside container-based applications can improve workload portability across cloud environments. Ansible should also be considered as part of the infrastructure automation toolset. While Terraform provisions and manage cloud infrastructure, Ansible complements it by automating configuration management, application deployment and orchestration. Together, both tools support an automated infrastructure lifecycle, reducing administrative effort while improving operational efficiency, consistency and governance.

(HashiCorp, 2026; RedHat, 2026).

2.3 Scalability and Cost Optimisation

Scalability in the Cloud allows CoinMeister to adjust its computing resources according to business requirements. Automatic Auto-scaling and elasticity, which can deliver additional required computing capacity during Increased transaction periods and reduce resources during quieter periods. This possibility improves service reliability in the way applications can maintain performance during times of increased customer activity. It is also helping reduce load and costs because CoinMeister avoids maintaining unnecessary infrastructure capacity. Using pay-as-you-use cloud model,

such models provide additional benefits in terms of cost efficiency. in teh pre-cloud era, businesses had to purchase large amounts of expansive and permanent hardware, involving extra CAPEX and OPEX costs, and also hardware disposal and electricity costs, the company pays only for resources consumed in the specific cloud model, reducing maintenance and administration costs and efforts at the datacenter, in between other levels. Resource monitoring tools help identify inefficient workloads and optimize storage, computing and application usage. Through effective resource management, CoinMeister can reduce operational expenditure (OPEX) and capital expenditure costs (CAPEX), while maintaining high-quality services.

2.4 Security Architecture

An effective security architecture is a fundamental and uncompromising requirement for CoinMeister, platforms of financial providers are crucially depending on implementation of robust cybersecurity capabilities that are accurately aligned with Governance, Risk and Compliance (GRC) requirements, and such that continuously monitoring evolving cyber threats. The cloud security architecture should focus Zero Trust philosophy and principles at the core of all life cycle, this includes strong encryption, identity and access management (IAM) with multi-factor authentication (MFA), Key Vault/Secrets Manager, Hardware Security Modules (HSM) where appropriate, Web Application Firewalls (WAF), DDoS protection, Security Information and Event Management (SIEM) monitoring, and backup and disaster recovery mechanisms. Strong encryption helps protect confidentiality during storage, use and transmission. IAM provides authorized access for users and applications, where everyone receives the least amount of permission required for their roles or operation

boundaries and this is strictly according to the principle of least privilege, while MFA provides an additional layer of protection against compromised credentials. The introduction of AI-based monitoring will help to improve the identification of suspicious behavior, using machine learning and deep learning techniques thus enhancing cloud security monitoring and improving threat detection capabilities.

(Alzoubi et al. 2024).

Effective backups and recovery capabilities are essential, especially for protecting against ransomware, system failures and accidental data loss. These mechanisms support the ability to maintain improved service availability while protecting customer information. Strong backups and recovery capabilities play a crucial role for CoinMeister to become and remain competitive.

(Snedaker, 2013).

3. Integration of Advanced Cloud Technologies

3.1 Artificial Intelligence Integration

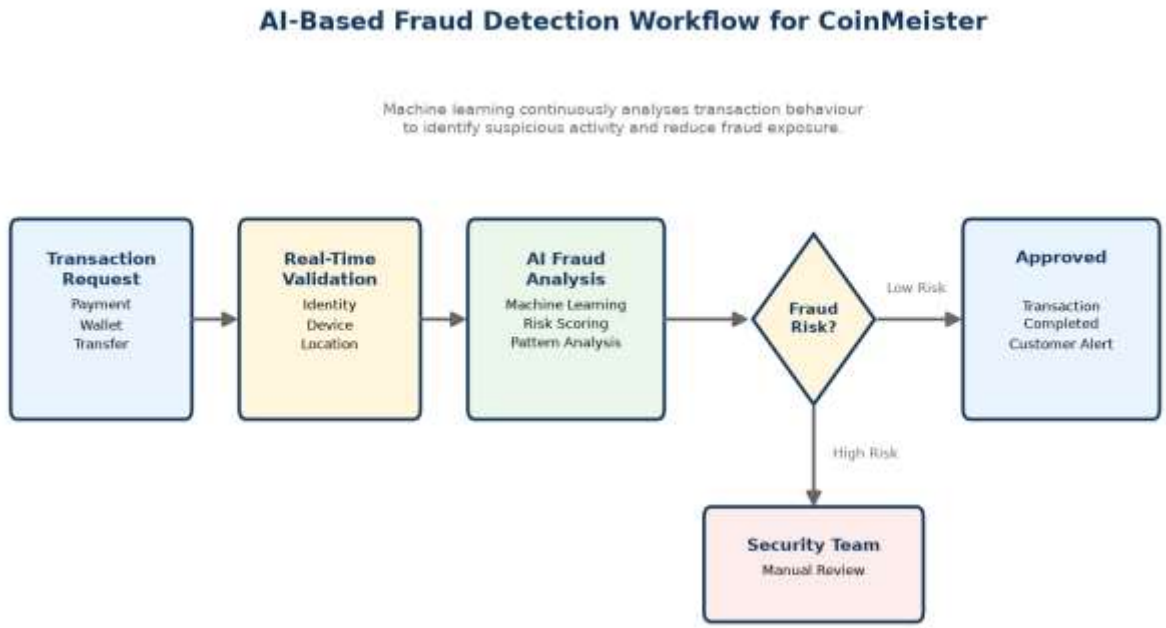
CoinMeister’s can utilize AI operations for improvement through fraud detection, data analysis, automated support and administration effort and cost. Using machine learning models, transaction analysis can be used to monitor patterns and identify suspicious activities, such as unusual locations, unexpected payment amounts or abnormal customer behavior. AI-driven fraud detection supports more rapid identification of potential threats compared with traditional manual monitoring and early detection is pivotal for prevention. Customer behavior analysis can also help CoinMeister deliver personalized services and improve customer-needs tailored

services. Modern cloud platforms possess the capabilities, tools and computing resources required to adopt advanced AI applications, with less dependency on in-house development team, allowing organizations to adopt AI capabilities without maintaining expensive specialized infrastructure.

(Carugati, 2023).

AI customer support systems could also provide benefit in service efficiency by handling common enquiries, reducing human-based support and providing faster responses.

Figure 2: AI Fraud Detection Workflow



3.2 Cloud Automation

Automation can support improved software development and operational efficiency. Using DevOps practices, those can be automated and integrate development, testing

and deployment phases, thus enabling faster and improved delivery of services. CI/CD pipelines can be used to automatically test software changes before deployment, hence reducing errors and improving reliability. Automated compliance checks can establish basis for applications to meet security baseline enforcement before release, while infrastructure monitoring automation would provide continuous insight into system performance, security related events and resource usage.

3.3 Serverless Computing

Cloud providers can automatically allocate resources as needed, making serverless suitable for event-driven workloads, where organizations can make use of serverless computing to allow applications to run without the need to directly manage servers. CoinMeister could use serverless computing for transaction notifications, API processing and customer alerts. For instance, completed transactions could automatically trigger confirmation notifications. However, serverless introduces risks including access control issues and application dependencies.

(Toosi et al., 2024; Ni et al., 2024).

3.4 Hybrid Cloud Benefits

As discussed in Section 2.1, Hybrid cloud provides CoinMeister with greater flexibility, resilience and compliance. Hybrid cloud separates sensitive financial workloads from scalable public cloud services while maintaining governance and flexibility, may lack the flexibility of Private cloud, those have advantages services and improved innovation. This approach allows CoinMeister to benefit from both.

4. Risk and Compliance Considerations

4.1 Cybersecurity Risks

As discussed in Section 2.4, Like many within the industry, CoinMeister faces cyber threats which include, but not limited to; ransomware, account compromise, data breaches system failures, third-party service disruptions, and much more. These threats could interrupt services, expose customer information and damage reputation. To reduce these risks to an acceptable level, it is highly important that CoinMeister implement improved governance, to control strong encryption, efficient IAM practices and continuous monitoring. AI security tools can identify suspicious activity and improve threat response capabilities.

(Alzoubi, et al. 2024).

Regular vulnerability assessments and penetration testing should also be conducted to identify security weaknesses.

4.2 Financial Compliance Risks

CoinMeister portfolio focuses mainly on digital payment and wallet services, therefore compliance with requirements such as PCI DSS, GDPR and AML/KYC obligations is essential to protect customer data and financial operations. Audit logging must multiple activities to be able to track events and ensure non-repudiation, system changes and transactions. Last privilege and role-based access management provide authorized employees with access only to information required for their responsibilities. Cloud security requires shared responsibility between providers and organizations. Cloud

providers protect infrastructure, CoinMeister remains responsible for applications, access management and customer data protection.

(Mather, et al. 2009; PCI Security Standards Council, 2024; European Union, 2016).

4.3 Business Continuity Risks

Financial services generally require high availability of critical services. CoinMeister should plan and implement disaster recovery with defined Recovery Time Objectives (RTO) and Recovery Point Objectives (RPO) based on business requirements in order to establish acceptable service restoration times and data loss limits for ensuring that disaster recovery measures are fit-for-purpose.

(Snedaker, 2013).

4.4 Vendor Lock-in Risk

Overseeing the risk of vendor-dependent architecture and vendor lock-in may restrict the organization’s ability to adapt, if CoinMeister becomes dependent on one cloud provider. Portable technologies, Infrastructure as Code and containerization can reduce this risk. A carefully tailored multi-cloud strategy can help to improve adaption, although it requires effective governance and operational management.

(Dunbar, 2024).

5. Future Recommendations

CoinMeister should prepare and evaluate edge computing in order to allow rapid Transaction processing and real-time fraud prevention. Processing data nearer to customers can reduce delays and improve transaction monitoring. The company

should also prepare quantum cloud products, because future quantum computing is very likely to pose a challenge in current encryption methods. Investment in quantum-resistant technology will help protect long-term business activities. Constant AI development and improved multi-cloud flexibility and maturity should remain priorities. These capabilities will help CoinMeister remain competitive and adaptable.

6. Conclusion

Choosing a hybrid cloud strategy could enable CoinMeister to work with a well-balanced foundation, a combination of secure-by-design and privacy focused and scalable flexibility for future growth. Combining private cloud control with public cloud flexibility improves innovation while maintaining compliance. IaC, automation, AI, serverless computing and advanced security controls will enhance resilience and operational efficiency. guided by effective governance, disaster recovery and cybersecurity practices will protect customer information and adopt emerging technologies such as edge computing and quantum-resistant security will support long-term competitiveness.

Figure 3: Business Value Chain Enabled by the Proposed Hybrid Cloud Strategy

Business Value Chain Enabled by the Proposed Hybrid Cloud Strategy



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